Memory Notes

Simplified LPDDR2 Simplified edition - JEDEC JESD209-2F

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1 Simplified Lpddr2 Index

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2 Introduction

2.1 What LPDDR2 SDRAM is

LPDDR2 SDRAM is the second generation of low-power double-data-rate memory, standardized by JEDEC as JESD209-2F. It is designed for mobile and battery-powered systems: the interface runs unterminated at 1.2 V, and the device carries a family of power-reduction features (per-bank refresh, partial-array self-refresh, deep power-down) that desktop DDR2/DDR3 do not have.

Its most unusual trait, relative to the desktop DDR family, is the command/address (CA) interface: instead of a wide SDR address bus with dedicated RAS_n/CAS_n/WE_n strobes, all commands and addresses travel on a 10-bit DDR bus (CA0-CA9), clocked in on both edges of the clock. One command occupies one clock cycle, two CA edges.

The spec covers three device classes; this book is about the SDRAM ones:

Class	What it is	Prefetch
LPDDR2-S2	SDRAM, 2n prefetch (lower speed grades)	2n
LPDDR2-S4	SDRAM, 4n prefetch (the mainstream part)	4n
LPDDR2-N	Non-volatile memory on the same bus (out of scope here)	4n

Family positioning, in one line each:

Tech	Command bus	Distinctives
LPDDR (JESD209)	SDR	First low-power DDR, PASR and temperature-compensated refresh arrive
LPDDR2 (JESD209-2)	10-bit DDR CA	Per-bank refresh, deep power-down, ZQ calibration, HSUL_12 IO
DDR2 (JESD79-2)	SDR	Desktop: ODT, posted CAS, DLL; none of which LPDDR2 has
LPDDR3 (JESD209-3)	10-bit DDR CA	Successor; same CA concept, faster

2.2 Key features of LPDDR2

- 10-bit double-data-rate CA bus: command and address packed onto two edges of one clock cycle. CS_n and CKE are the only SDR control pins.
- 4n prefetch on S4 (2n on S2): an S4 column access moves 4 words internally per DQ per access.
- Burst lengths 4, 8 or 16; sequential or interleaved order; wrap or no-wrap (no-wrap exists only at BL4).
- No DLL. Read data timing is specified as tDQSCK from the clock, and tDQSCK may span multiple clock periods.
- No ODT. The IO is HSUL_12 (high-speed unterminated logic, 1.2 V), so bus turnarounds are contention-avoidance problems, not ODT switching problems.
- No bank groups. Banks are a flat set of 4 or 8.
- No DBI, no parity, no on-die ECC. Data integrity is entirely the controller's problem.
- Per-bank refresh (REFpb) on 8-bank devices, in addition to all-bank refresh (REFab).
- Partial Array Self-Refresh (PASR): refresh only selected banks or segments during self-refresh.
- Deep Power-Down (DPD): lowest power state; array contents are lost.
- Temperature sensor readable via MR4, used for refresh-rate scaling and AC timing de-rating (temperature-compensated self refresh).
- ZQ output-driver calibration (S4 and N only; S2 ignores it).
- Densities 64 Mb to 8 Gb (SDRAM), organizations x8, x16, x32.

2.3 Speed grades

Speed grades are named by data rate: LPDDR2-1066, -933, -800, -667, -533, -466, -400, -333, -266 and -200, corresponding to clock frequencies of 533 MHz down to 100 MHz. Core timing (tRCD, tRP, tRAS) is specified in nanoseconds with fast/typical/slow bins, not per-grade clock counts.

2.4 The simplified drill model

Real parts have 8K-16K rows and up to 2K columns per row. All examples in this book instead use a deliberately tiny geometry so bank state is obvious at a glance:

- Banks: B0-B7 (eight, as on 1 Gb and larger parts)
- Rows: R0-R7 per bank
- Columns: C0-C7 per row

This model is used by the ddr_drills training app. Every command trace, walkthrough and timing example in Chapters 3, 4 and 6 uses it. Real address maps are far larger; the scheduling rules are identical.

Source: JESD209-2F sections 1, 3, Table 103

3 Organization

3.1 Banks, rows, columns

An LPDDR2-SX device is organized as 4 or 8 banks of independent DRAM arrays. Each bank has its own sense-amplifier row buffer, so up to 4 or 8 rows can be open simultaneously (one per bank). Density determines the bank count: 64 Mb through 512 Mb parts have 4 banks, 1 Gb and larger parts have 8 banks (S2/S4; check the addressing table for exact splits).

Address geometry per density (SDRAM, from the spec addressing table):

Density	Banks	Rows	Columns (x16)	tREFI (us, <=85 C)
64 Mb	4	R0-R11	C0-C7	15.6
128 Mb	4	R0-R11	C0-C8	15.6
256 Mb	4	R0-R12	C0-C8	7.8
512 Mb	4	R0-R12	C0-C9	7.8
1 Gb	4 (S2) / 8 (S4)	R0-R13 / R0-R12	C0-C9	7.8
2 Gb	4 (S2) / 8 (S4)	R0-R14 / R0-R13	C0-C9	3.9
4 Gb	8	R0-R13	C0-C10	3.9
6 Gb	8	R0-R14	C0-C10	3.9
8 Gb	8	R0-R14	C0-C10	3.9

x8 parts gain one column bit, x32 parts lose one, relative to x16.

Two addressing rules to remember:

- C0 is never transmitted on the CA bus; it is implied zero. Column addresses on the bus start at C1.
- 6 Gb parts have no memory where R13 and R14 are both high; an ACT or write to that space is ignored.

3.2 Prefetch and IO width

- S4: 4n prefetch. A column access fetches 4 words per DQ internally; the interface delivers 2 words per DQ per clock (DDR), so a minimum burst occupies 2 clocks. This is why tCCD = 2 clocks on S4.
- S2: 2n prefetch, 1 clock minimum per burst, tCCD = 1 clock.

Data bus widths are x8, x16 and x32. Each byte lane has its own differential strobe (DQS_t/DQS_c) and its own DM mask pin: one DQS pair and one DM per 8 DQs.

3.3 Burst structure

Accesses are burst-oriented: one Read or Write command moves BL words per DQ, starting at the issued column and following the programmed burst order (sequential or interleaved, wrap or no-wrap). BL is 4, 8 or 16, programmed in MR1.

3.4 Channels

Multi-channel parts exist at the package level: a PoP package can carry two fully independent LPDDR2 channels (each with its own CA bus, clock, CKE and CS_n). Channels share nothing but the package; the protocol in this book is per-channel.

3.5 The simplified drill model

Per Chapter 1, all examples use 8 banks (B0-B7), 8 rows (R0-R7) and 8 columns (C0-C7). Real parts are far larger; the small model exists to make bank/row state obvious in command traces.

Source: JESD209-2F sections 2.13 (Table 3), 3

4 Acronyms and Conventions

Terms used precisely in this book.

4.1 Signal names

Term	Meaning
CK_t / CK_c	Differential clock. Positive edge = CK_t rising through CK_c falling
CKE	Clock enable. SDR, sampled on the positive clock edge; gates all power states
CS_n	Chip select. SDR, sampled on the positive clock edge; part of the command code
CA0-CA9	10-bit DDR command/address bus; sampled on both clock edges
CAxr / CAxf	CA bit x on the rising (r) / falling (f) edge of the clock
DQ	Bidirectional data
DQS_t / DQS_c	Differential data strobe, one pair per byte lane; edge-aligned on reads, center-aligned on writes
DM	Data mask, one per byte lane; masks write data when sampled high
ZQ	Reference pin for output drive-strength calibration

4.2 Commands

Term	Meaning
ACT	Activate (open a row in a bank)
RD / WR	Burst read / burst write
PRE	Precharge (close a row); per-bank or all-bank via the AB flag
REFab / REFpb	Refresh, all banks / one per-bank-refresh bank
BST	Burst terminate
MRW / MRR	Mode register write / read
NOP	No operation (two legal encodings)
SREF / PD / DPD	Self-refresh, power-down, deep power-down

4.3 Architecture terms

Term	Meaning
Prefetch (4n/2n)	Words moved internally per column access; S4 = 4n, S2 = 2n
BL / BT / WC	Burst length / burst type (sequential, interleaved) / wrap control (wrap, no-wrap)
RL / WL	Read latency / write latency, in clocks, programmed in MR2
nWR	Write recovery in clocks for auto-precharge, programmed in MR1
AP	Auto-precharge flag (CA0f of a RD/WR command)
AB	All-bank flag (CA4r of a PRE command)
RU{}	Round-up function; RU{x} = smallest integer >= x

4.4 Power and refresh terms

Term	Meaning
PASR	Partial array self-refresh: refresh only masked banks/segments in self-refresh
TCSR	Temperature-compensated self refresh (refresh rate scaled by temperature, via MR4)
DAI	Device auto-initialization status bit (MR0 OP0)
TUF	Temperature update flag (MR4 OP7)
DNV	Data-not-valid (an NVM feature; SDRAM does not implement it)
HSUL_12	High-speed unterminated logic, 1.2 V: the LPDDR2 IO standard
tREFW / tREFI	Refresh window / average refresh interval
tRFCab / tRFCpb	Refresh cycle time, all-bank / per-bank

4.5 Conventions

- H/L in truth tables are logic high/low at the sampled edge; X is “don’t care (but a defined logic level)”.
- Timing symbols are written as plain text: tRCD, tWTR, tREFIpb.
- Anything the spec leaves unspecified is illegal; after an illegal event the device must be powered down and re-initialized.

Source: JESD209-2F sections 2.12 (Table 2), 5.18 (notes to Table 60)

5 Power-Up and Initialization

LPDDR2 must be powered up and initialized by a fixed sequence; any other procedure is undefined behavior. The sequence below paraphrases the mandatory steps. Named checkpoints (Ta..Tg) are the spec's.

5.1 Cold start sequence

1. **Power ramp (Ta -> Tb).** Apply supplies with CKE held low (<= 0.2 x VDDCA). Ta is when any supply first reaches 300 mV; Tb is when all supplies and references are in range. The ramp (tINIT0) must complete within 20 ms. Ordering rules during the ramp: VDD1 within 200 mV above VDD2, VDD1/VDD2 within 200 mV above VDDCA and VDDQ, VREF always below every supply, ground pins within 100 mV of each other.
2. **CKE and clock (Tb -> Tc).** Keep CKE low for at least tINIT1 = 100 ns after Tb. The clock must be stable for at least tINIT2 = 5 tCK before CKE's first low-to-high transition. If any mode-register reads will be done during init, the clock must be in the boot range tCKb = 18-100 ns (about 10-55 MHz).
3. **NOP wait (Tc -> Td).** With CKE high, issue NOP for at least tINIT3 = 200 us.
4. **Reset (Td).** Issue MRW Reset (an MRW to MR63; any operand). Then hold CKE high with NOPs for at least tINIT4 = 1 us.
5. **DAI polling (Td -> Tf).** Only MRR and power-down entry/exit are legal now. Poll MR0's DAI bit (or simply wait tINIT5 = 10 us for SDRAM) until the device reports auto-initialization complete; the device is then Idle. This is also when MRRs of MR0/MR5/MR8 identify device type, manufacturer, density and IO width.
6. **ZQ calibration (Tf -> Tg).** Issue MRW ZQ Initialization Calibration (MR10, code 0xFF). The device is ready for normal operation after tZQINIT (1 us). On multi-device buses, ZQ calibrations must not overlap.
7. **Configure.** MRW the operating registers: MR1 (burst length, burst type, wrap, nWR), MR2 (RL/WL), MR3 (drive strength). The device is now Idle and accepts any valid command.

5.1.1 Figure 2.1: LPDDR2 Power-Up Initialization Timeline

LPDDR2 power-up initialization: checkpoints Ta to Tg

interval	Ta-Tb	Tb-Tc	Tc-Td	Td	Td-Te	Te-Tf	Tf-Tg	after Tg
phase	**power ramp**	**CKE low, clock on**	**NOP wait**	**MRW RESET**	**NOP hold**	**DAI poll**	**ZQ init**	**configure**
supplies	ramping	in range						
CK	x	stable	running					
CKE	0		**1**					
command	-	-	NOP	MRW MR63	NOP	MRR MR0 (DAI)	MRW MR10 = 0xFF	MRW MR1,2,3
timing	tINIT0 <= 20 ms	tINIT1 >= 100 ns tINIT2 >= 5 tCK	tINIT3 >= 200 us	-	tINIT4 >= 1 us	tINIT5 <= 10 us	tZQINIT >= 1 us	-

CKE stays at or below 0.2 x VDDCA through the ramp. The clock must be stable for tINIT2 before CKE's first rising edge; if MRRs will be issued during init it must sit in the boot range tCKb = 18-100 ns. Between Te and Tf only MRR and power-down entry/exit are legal; waiting out tINIT5 instead of polling DAI is allowed.

LPDDR2 Power-Up Initialization Timeline

5.2 Reset without power ramp

An MRW Reset issued later (outside power-up) restarts initialization at step 4: reset the mode registers to defaults, wait tINIT4, poll DAI or wait tINIT5, redo ZQ calibration and the MRW configuration. SDRAM array contents are undefined after MRW Reset.

5.3 Power-off

Controlled power-off mirrors the ramp rules in reverse: hold CKE low while supplies fall, keep all inputs between their rails to avoid latch-up, and complete the fall (Tx -> Tz, all supplies below 300 mV) within tPOFF = 2 s. The same supply-ordering inequalities apply during the fall.

Uncontrolled power-off is tolerated only within tightened bounds the spec gives separately; the safe design rule is that a system that cannot guarantee the controlled sequence must treat DRAM contents as lost.

5.4 Notes that bite

- Mode-register contents have defaults only after Device Auto-Initialization; before the MRW Reset they are undefined.
- MR1, MR2 and MR3 must all be written even if the defaults look right - the whole flow assumes it.
- Boot-frequency rules (tCKb, tISb, tIHb, tDQSCKb) apply until the device is configured; some AC parameters are relaxed before then.
- After DPD exit, the full power-up sequence from step 3 applies.

Source: JESD209-2F sections 3.4.1-3.4.4, 5.13.3

6 Mode Registers

LPDDR2 replaces the MRS/EMRS command-address encoding of desktop DDR with two dedicated commands: MRW (write) and MRR (read), each carrying an 8-bit register address (MA0-MA7) on the CA bus. Mode registers are 8 bits wide (OP0-OP7). The table below covers every register that matters for SDRAM operation; the NVM-only registers (MR20-MR31) are omitted.

6.1 Register map (SDRAM view)

MR	Name	Access	Contents
MR0	Device Info	R	DAI (init done), DI (SDRAM vs NVM), DNVI, RZQI
MR1	Device Feature 1	W	BL, BT, WC, nWR
MR2	Device Feature 2	W	RL & WL
MR3	I/O Config 1	W	DS (drive strength)
MR4	Refresh Rate / Temp	R	Refresh-rate code, TUF flag
MR5	Basic Config 1	R	Manufacturer ID
MR6 / MR7	Basic Config 2/3	R	Revision ID1 / ID2
MR8	Basic Config 4	R	I/O width, density, type (S2/S4/N)
MR9	Test Mode	W	Vendor-specific test mode
MR10	IO Calibration	W	ZQ calibration codes
MR16	PASR Bank Mask	W	Self-refresh bank masking (S2 ranges; S4 per-bank bits)
MR17	PASR Segment Mask	W	Segment masking (1 Gb+ S4 only)
MR32 / MR40	DQ Cal Pattern A/B	R	Read-only training patterns
MR63	Reset	W	Any write = MRW Reset

Rules: RFU bits are written 0 and read as 0; writes to read-only registers have no effect; reserved registers must not be written.

6.2 MR1 - Device Feature 1 (the one you always program)

Field	Bits	Encodings
BL	OP2-0	010 = BL4 (default), 011 = BL8, 100 = BL16
BT	OP3	0 = sequential (default), 1 = interleaved
WC	OP4	0 = wrap (default), 1 = no-wrap (BL4 only)
nWR	OP7-5	001..110 = nWR 3..8 (default 3); RU(tWR/tCK)

nWR tells the device when to start the internal precharge after a write with auto-precharge (AP=1). Program it to RU(tWR/tCK).

6.3 MR2 - RL & WL

OP3-0	RL / WL	OP3-0	RL / WL
0001	3 / 1 (default)	0100	6 / 3
0010	4 / 2	0101	7 / 4
0011	5 / 2	0110	8 / 4

Pick the pair whose RL supports the target frequency (see the speed table in Chapter 6).

6.4 MR3 - Drive strength (DS)

OP3-0	Pull-down (typ)
0001	34.3 ohm
0010	40 ohm (default)
0011	48 ohm
0100	60 ohm
0110	80 ohm
0111	120 ohm (optional)

6.5 MR4 - Refresh rate and temperature (read-only)

- OP2-0: the device's recommended refresh rate for the current temperature: 4x, 2x, 1x (<= 85 C), 0.25x (with or without AC de-rating), or a temperature-limit-exceeded code. This is the temperature-compensated self-refresh (TCSR) mechanism: the controller scales tREFI/tREFW by this factor.
- OP7 (TUF): set when OP2-0 changed since the last MR4 read; reading MR4 clears it.
- When OP2-0 requests de-rating, add 1.875 ns to tRCD, tRC, tRAS, tRP and tRRD (and de-rate tDQSCK per the AC table).

6.6 MR10 - ZQ calibration codes

Code	Command	Latency
0xFF	Initialization calibration (after power-up)	tZQINIT = 1 us
0xAB	Long calibration	tZQCL = 360 ns
0x56	Short calibration	tZQCS = 90 ns
0xC3	ZQ reset (default calibration)	tZQRESET = 50 ns

S2 devices ignore ZQ commands. If ZQ is tied to VDDCA the device runs default calibration and ignores ZQ commands entirely.

6.7 MR16 / MR17 - PASR masks

- S2: MR16 OP1-0 selects full / 1/2 / 1/4 / 1/8 array, always anchored at bank 0 (lower banks keep refresh).
- S4: MR16 OP7-0 is a per-bank mask; 1 blocks refresh to that bank.
- S4, 1 Gb and up: MR17 OP7-0 is a per-segment mask (8 segments, each an eighth of the row space of every bank); 1 blocks refresh to that segment.

Masked regions lose data in self-refresh. See Chapter 3 (self-refresh) and Chapter 6 (refresh timing).

Source: JESD209-2F section 3.5 (Tables 17-22), 5.13.4

7 The DDR CA Interface

LPDDR2 has no RAS_n/CAS_n/WE_n and no wide address bus. Everything - command, bank, row, column, mode-register address and data - travels on CA0-CA9, clocked in on BOTH edges of one clock cycle. Understanding this packing is the key to the whole protocol.

7.1 One command = one clock = two edges

- Each command occupies exactly one clock cycle: ten CA bits on the rising edge (written CA0r..CA9r) and ten on the falling edge (CA0f..CA9f).
- CS_n and CKE are single-data-rate: they are sampled only on the rising edge.
- The command itself is decoded from CKE, CS_n and CA0r-CA3r (with CKE of the previous cycle for power-state entries). CA4r and above, and the whole falling edge, carry payload (addresses, bank, MR fields).
- CS_n high on the rising edge = Deselect (a form of NOP), regardless of the CA pins.

7.2 Command codes (rising edge)

CA0r CA1r CA2r CA3r	Command
L L L L	MRW
L L L H	MRR
L L H L	Refresh, per bank (REFpb)
L L H H	Refresh, all banks (REFab)
L H R8 R9	ACT (CA2r-CA6r carry row bits)
H L L x	WR
H L H x	RD
H H L L	BST (burst terminate)
H H L H	PRE (CA4r = AB all-bank flag)
H H H x	NOP

With CKE falling at this edge, three of these codes become power-state entries instead: REF code -> self-refresh entry; PRE code -> deep power-down entry; CS_n high (or NOP) -> power-down entry.

7.3 Where the address bits live

Per command, rising/falling CA assignments (x = don't care):

Command	Rising edge CA0-CA9	Falling edge CA0-CA9
ACT	L H R8 R9 R10 R11 R12 BA0 BA1 BA2	R0 R1 R2 R3 R4 R5 R6 R7 R13 R14
RD / WR	H L cmd x x C1 C2 BA0 BA1 BA2	AP C3 C4 C5 C6 C7 C8 C9 C10 C11
PRE	H H L H AB x x BA0 BA1 BA2	x x x x x x x x x x
MRW	L L L L MA0 MA1 MA2 MA3 MA4 MA5	MA6 MA7 OP0 OP1 OP2 OP3 OP4 OP5 OP6 OP7
MRR	L L L H MA0 MA1 MA2 MA3 MA4 MA5	MA6 MA7 x x x x x x x x

(cmd = H for RD, L for WR; x = don't care)

Consequences:

- A full row address (up to R0-R14) plus bank fits one ACT: upper row bits and bank on the rising edge, lower row bits and the top row bits (R13, R14) on the falling edge.
- A column address (C1-C11) plus bank plus the auto-precharge flag fits one RD/WR: C1, C2 and bank on the rising edge, AP and C3-C11 on the falling edge. C0 is implied zero and never transmitted.
- AP (auto-precharge) lives at CA0f of RD/WR. AP=1 closes the row when the burst completes.
- The mode-register address is split across edges: MA0-MA5 rising, MA6-MA7 falling. MRW data (OP0-OP7) rides CA2f-CA9f.

7.4 Why this matters for a controller

- The CA bus is a scarce resource: every command costs a full clock, including NOPs, so scheduling is about packing useful commands into every cycle the timing rules allow.
- Bank bits move around per command (CA7r-CA9r on ACT/RD/WR/PRE), and the command decoder needs both edges before it knows the full address.
- Because CKE/CS_n are SDR, power-state transitions align to rising edges only; the CKE truth table (Chapter 3) is the authority for what a CKE edge means.

Source: JESD209-2F section 5.18.1 (Table 60 and notes), 5.1, 5.4, 5.5

8 Command Truth Tables

Two tables govern what the device does: the command truth table (what the CA pins mean this cycle) and the CKE table (what a CKE edge means given the device's current state). Both are paraphrased below.

8.1 Command truth table

CKE(n-1) is CKE at the previous rising edge; CKE(n) and CS_n and CAxr are sampled at the current rising edge; CAxf at the current falling edge. H/L = logic level, X = don't care (but driven).

Command	CKE(n-1)	CKE(n)	CS_n	CA0r	CA1r	CA2r	CA3r	Falling edge payload
MRW	H	H	L	L	L	L	L	MA6 MA7, OP0-OP7
MRR	H	H	L	L	L	L	H	MA6 MA7
REFpb	H	H	L	L	L	H	L	X
REFab	H	H	L	L	L	H	H	X
Enter self-refresh	H	L	L	L	L	H	X	X
ACT	H	H	L	L	H	R8	R9	R0-R7, R13, R14
WR	H	H	L	H	L	L	X	AP, C3-C11
RD	H	H	L	H	L	H	X	AP, C3-C11
PRE (bank/all)	H	H	L	H	H	L	H	X
BST	H	H	L	H	H	L	L	X
Enter DPD	H	L	L	H	H	L	X	X
NOP	H	H	L	H	H	H	X	X
Deselect	H	H	H	X	X	X	X	X

Command	CKE(n-1)	CKE(n)	CS_n	CA0r	CA1r	CA2r	CA3r	Falling edge payload
(NOP)								
Enter power-down	H	L	H	X	X	X	X	X
Exit PD/SREF/DPD	L	H	H	X	X	X	X	X
Maintain PD/SREF/DPD	L	L	X	X	X	X	X	X

Notes that load-bear:

- ACT rising edge: CA2r-CA6r = R8-R12, CA7r-CA9r = BA0-BA2; falling edge CA0f-CA7f = R0-R7, CA8f = R13, CA9f = R14.
- RD/WR rising edge: CA5r = C1, CA6r = C2, CA7r-CA9r = BA0-BA2; falling edge CA0f = AP, CA1f-CA9f = C3-C11.
- PRE rising edge: CA4r = AB (1 = all banks, bank address ignored), CA7r-CA9r = BA0-BA2 for the per-bank case.
- MRW/MRR rising edge: CA4r-CA9r = MA0-MA5; falling edge CA0f = MA6, CA1f = MA7; MRW falling edge CA2f-CA9f = OP0-OP7.
- REFpb exists only on 8-bank devices.
- Self-refresh and DPD exits are asynchronous (CKE rise); power-down exit is synchronous.

8.2 CKE table (power-state transitions)

Current state	CKE(n-1)	CKE(n)	CS_n	Command	Next state
Active power-down	L	L	X	X	Active power-down
Active power-down	L	H	H	NOP	Active (exit, wait tXP)
Idle power-down	L	L	X	X	Idle power-down
Idle power-down	L	H	H	NOP	Idle (exit, wait tXP)
Self refresh	L	L	X	X	Self refresh
Self refresh	L	H	H	NOP	Idle (exit, wait tXSR)
Deep power-down	L	L	X	X	Deep power-down
Deep power-down	L	H	H	NOP	Power on (full re-init)
Banks active	H	L	H	NOP	Active power-down
All banks idle	H	L	H	NOP	Idle power-down
All banks idle	H	L	L	REF code	Self refresh
All banks	H	L	L	DPD	Deep

Current state	CKE(n-1)	CKE(n)	CS_n	Command	Next state
idle				(PRE) code	power-down
Any (CKE high)	H	H	-	per command truth table	-

The clock must toggle at least twice during tXP and during tXSR. States and sequences not shown are illegal.

8.3 NOP encodings

NOP has two legal forms: CS_n high at the rising edge (Deselect), or CS_n low with CA0r = CA1r = CA2r = H. A NOP never terminates an in-flight operation; it just keeps the bus quiet. NOP is the mandatory follow-up to every power-state entry and exit.

Source: JESD209-2F sections 5.17, 5.18.1 (Table 60), 5.19 (Table 61)

9 Commands: Rules and Constraints

Each command below: what it does, its encoding, and what may follow. Timing symbols are defined in Chapter 6.

9.1 ACT - Activate

Opens row R of bank B (sense amps latch the row). Required before any RD/WR to that row.

- Encoding: CA0r=L, CA1r=H; bank on CA7r-CA9r, row split across edges.
- RD/WR to the same bank may follow after tRCD.
- ACT to a different bank may follow after tRRD.
- ACT to the same bank again requires a PRE first; the ACT-PRE-ACT round trip must satisfy tRAS (ACT to PRE) and tRC (ACT to ACT).
- 8-bank devices only: no more than 4 ACTs in a rolling tFAW window, and REFpb counts as an activation for tFAW.

9.2 RD / WR - Burst read / write

Starts a burst of BL words at column C of the open row in bank B.

- Encoding: CA0r=H, CA1r=L, CA2r = H (RD) / L (WR); AP at CA0f.
- AP=1: the bank auto-precharges after the burst (internal precharge starts after RL/tRAS rules for reads, after nWR for writes).
- Read data appears RL*tCK* + *tDQSCK* + *tDQSQ after the command's rising edge; write data is expected WL*tCK + tDQSS after the command.
- Another RD (after RD) or WR (after WR) may follow after tCCD (2 clocks on S4, 1 on S2). On S4 with BL4 a burst may NOT be interrupted; with BL8/BL16 interruption is allowed only on even clock cycles after the first command.
- WR after RD: at least RL + RU(tDQSCKmax/tCK) + BL/2 + 1 - WL clocks after the RD command (bus turnaround; see Chapter 6).
- RD after WR: at least WL + 1 + BL/2 + RU(tWTR/tCK) clocks after the WR command.
- PRE to the same bank may follow a RD after tRTP (min), a WR after tWR; tRAS must also be met.

9.3 PRE - Precharge

Closes the open row of one bank (AB=0) or all banks (AB=1).

- Encoding: CA0r=H, CA1r=H, CA2r=L, CA3r=H; AB at CA4r.
- The bank is ready for a new ACT after tRPpb (single-bank) or tRPab (all-bank). On 8-bank devices tRPab is 3 ns longer; on 4-bank devices they are equal.
- May not be issued before tRAS after ACT, tRTP after RD, or tWR after WR (to that bank).

9.4 REFab / REFpb - Refresh

REFab refreshes all banks (all must be idle); REFpb refreshes one bank, chosen by the device's internal round-robin counter (0-1-…-7-0-…).

- Encoding: CA0r=L, CA1r=L, CA2r=H; CA3r = H (all) / L (per bank).
- REFpb is 8-bank devices only; the target bank must be idle, and the controller must track the round-robin position (it resets to bank 0 on reset, on self-refresh exit, and on every REFab).
- After REFab: wait tRFCab before any ACT or further REF. After REFpb: wait tRFCpb before an ACT to the same bank or another REF; only tRRD before an ACT to a different bank.
- During REFpb the other banks stay fully usable - that is its purpose.

9.5 BST - Burst terminate

Truncates the most recent RD or WR burst.

- Encoding: CA0r=H, CA1r=H, CA2r=L, CA3r=L.
- Legal only within BL/2 - 1 clocks after the RD/WR command (S4); on S2 it may be issued in any cycle after the command.
- Effective burst length after BST = 2 x (clocks from RD/WR to BST). Use the effective BL in all subsequent turnaround math.
- The truncation takes effect one full read/write latency later.

9.6 MRW / MRR - Mode register write / read

- MRW: all banks must be idle; no command other than NOP for tMRW (5 clocks) after. Any MRW to MR63 is a device Reset.
- MRR: fixed BL4; register contents appear on DQ0-DQ7 of the first beat only (later beats undefined); DQS toggles for the whole burst; the operation may not be interrupted; tMRR = 2 clocks between commands.
- MRR may not be issued earlier than BL/2 after a RD, or WL + 1 + BL/2 + RU(tWTR/tCK) after a WR.

9.7 Self-refresh entry / exit

- Entry: CKE falls with the REF encoding, all banks idle. The device refreshes itself from an internal timer; all inputs except CKE become don’t-care; VDDQ may be powered down.
- Minimum stay tCKESR; the clock may be stopped or re-tuned one clock after entry, but must be stable 2 clocks before exit.
- Exit: CKE rises (asynchronous); NOPs until tXSR (= tRFCab + 10 ns) expires. At least one refresh (1 REFab or 8 REFpb) must be issued after exit before the next self-refresh entry.

9.8 Power-down (PD)

- Entry: CKE falls with CS_n high (or NOP). With all banks idle it is idle PD; with a row open it is active PD. Input/output buffers (except CK and CKE) shut down; no refreshes occur, so dwell time is bounded by the refresh requirement.
- Minimum stay tCKE; exit on CKE rise with CS_n high; first valid command after tXP.

9.9 Deep power-down (DPD)

- Entry: CKE falls with the PRE encoding, all banks idle. Nearly everything inside the device powers down; array contents may be lost.
- Minimum stay tDPD = 500 us. Exit on CKE rise, then the full initialization sequence (from the tINIT3 NOP wait) is mandatory.

9.10 NOP / Deselect

Keeps the bus idle; never aborts anything. Mandatory after power-state entries/exits and wherever no other command is legal.

Source: JESD209-2F sections 5.1, 5.3-5.7, 5.9-5.15, 5.17

10 Walkthrough: AXI Read, Row Already Open

Configuration used in all Chapter 4 traces (the drill model): S4 device, BL = 4, RL = 3, WL = 1, sequential/wrap bursts. Banks B0-B7, rows R0-R7, columns C0-C7.

10.1 Scenario

An AXI4 master issues a 32-bit read (ARLEN = 3 on a 32-bit bus, i.e. 4 beats = 16 bytes) at an address that maps to bank B3, row R5, column C2. Row R5 is already open in B3 from an earlier access.

10.2 Command trace

```
RD  B3 C2          ; row already open - issue the burst immediately
NOP                ; tCCD = 2 clocks minimum between bursts (bus stays
quiet)
NOP                ; read data returns RL*tCK + tDQSCK + tDQSQ after RD
```

10.3 Why each command

- `RD B3 C2` - the row is open, so no ACT is needed. CA2r=H marks a read; C2 is the burst start column; AP=0 (CA0f) so the row stays open for the next access.
- The burst delivers 4 words (BL=4) on each DQ edge-aligned to DQS, starting RL clocks (plus tDQSCK) after the command.
- No PRE: leaving the row open is the open-page policy; a later access to R5 in B3 is again a hit.

10.4 AXI view

- AR channel: one 16-byte burst request.
- The DRAM burst of 4 words x (bus width) covers it exactly at BL=4 on a 32-bit DQ interface. A wider AXI beat count or narrower DRAM bus would need multiple RD commands spaced tCCD apart, e.g. `RD B3 C2`, `RD B3 C4` two clocks later, data returning back-to-back.

10.5 What to notice

- A row hit costs exactly one CA cycle on the command bus. Every other scheduling rule (tCCD, turnaround) is about packing the next command around it.
- The controller knew this was a hit because it tracks one open row per bank - that bookkeeping is the whole game.

Source: JESD209-2F sections 5.3, 5.4 (derived walkthrough)

11 Walkthrough: AXI Read, Wrong Row Open

11.1 Scenario

An AXI4 read request maps to bank B3, row R6, column C0 - but row R5 is currently open in B3. This is a row conflict (miss): the open row must be closed before the new one can be opened.

11.2 Command trace

```
PRE B3           ; close the wrong row (tRAS since its ACT already
met)
--- (tRPpb) ---  ; precharge time: bank unavailable
ACT B3 R6        ; open the requested row
--- (tRCD) ---   ; row-to-column delay
RD  B3 C0        ; burst read, AP=0, row stays open
NOP
NOP
```

11.3 Why each command

- `PRE B3` - a bank holds exactly one open row. AB=0 (CA4r), so only B3 is precharged; every other bank is untouched.
- `--- (tRPpb) ---` - single-bank precharge time (typ 18 ns). A PRE to one bank used tRPpb; a PRE-all would have cost tRPab (3 ns more on 8-bank parts) and closed banks that were doing useful work.
- `ACT B3 R6` - opens the requested row. The ACT-to-ACT spacing to B3 (prior ACT -> PRE -> this ACT) satisfies tRC = tRAS + tRPpb by construction if tRAS was met before the PRE.
- `--- (tRCD) ---` - row-to-column delay (typ 18 ns); the column circuitry cannot be driven until the row's data is in the sense amps.
- `RD B3 C0` - the read itself, identical to the hit case.

11.4 AXI view

Same single 16-byte AR request as the hit walkthrough - but the latency the master sees is now tRPpb + tRCD longer. Nothing on the AXI side changes; the penalty is entirely in DRAM scheduling.

11.5 What to notice

- Miss cost = tRPpb + tRCD (typ 36 ns at typical bin) before the first data can even start its RL journey.
- If the access had been issued with AP=1 on the PREVIOUS access to B3, the bank would already be idle and only tRCD would be owed - that is the close-page trade in one line.
- PRE may not be issued earlier than tRAS after the ACT that opened R5, nor earlier than tRTP after a read from R5. A scheduler that wants the earliest PRE starts the tRAS/tRTP clocks at the ACT/RD, not at the miss.

Source: JESD209-2F sections 5.1, 5.9 (derived walkthrough)

12 Walkthrough: AXI Write to an Idle Bank

12.1 Scenario

An AXI4 write (AW + 4 W beats, 16 bytes) maps to bank B1, row R2, column C0. B1 is idle (no open row).

12.2 Command trace

```
ACT B1 R2          ; open the row (bank was idle - no PRE needed)
--- (tRCD) ---     ; row-to-column delay
WR  B1 C0          ; burst write, AP=0; write data starts WL*tCK + tDQSS
later
NOP                ; burst occupies the DQ bus; tWR recovery runs from
                   ; the last data, then the bank may be precharged
```

12.3 Why each command

- `ACT B1 R2` - an idle bank needs no precharge; ACT is legal as long as tRRD from the previous ACT (any bank) and the tFAW window (max 4 ACTs per rolling tFAW, 8-bank parts) are met.
- `--- (tRCD) ---` - same row-to-column delay as a read.
- `WR B1 C0` - CA2r=L marks a write. The controller drives DQS with a tWPRE preamble and the first data WL clocks (plus tDQSS) after the command. Each beat must meet tDS/tDH around the DQS edge; DM pins mask any beats that should not be written.
- The trailing NOPs cover tWR (write recovery, typ 15 ns from the end of the burst data): the bank may not be precharged until tWR has elapsed, because the write is still being committed to the array.

12.4 AXI view

- AW and W channels carry the request; the DRAM does not see them separately - the controller issues WR only when it has (or is about to have) the write data, since LPDDR2 gives it no way to stall mid-burst.
- B (write response) can be returned once the data is accepted by the DRAM schedule; the tWR window is a DRAM-internal matter.

12.5 What to notice

- An idle-bank write miss costs tRCD, half the read-miss penalty - no tRP because nothing was open.
- Alternative: issue `WR B1 C0` with AP=1. The device then auto-precharges after nWR clocks, and the bank returns to idle with no explicit PRE - at the price of having programmed nWR = RU(tWR/tCK) correctly in MR1.
- tWR is the write-side analog of tRTP on the read side: both gate the PRE; both exist because the burst's last data and the array's precharge cannot overlap.

Source: JESD209-2F sections 5.1, 5.5, 5.9.2-5.9.3 (derived walkthrough)

13 Walkthrough: Mixed Reads and Writes

13.1 Scenario

Three AXI requests arrive close together:

1. Read: bank B3, row R5 (open), column C2.
2. Write: bank B1, row R2 (open), column C0.
3. Read: bank B1, row R2, column C4.

Both rows are already open, so this trace is pure bus scheduling: the DQ bus is shared, and read/write direction changes cost bubbles.

13.2 Command trace

```
RD  B3 C2         ; read burst 1: data out after RL + tDQSCK
--- (turnaround) ---  ; read-to-write bubble, see below
WR  B1 C0         ; write burst: DQ direction flips
--- (tWTR) ---    ; write-to-read recovery before any new RD
RD  B1 C4         ; read burst 2, same open row as the write - a hit
PRE B1            ; done with B1 (tRTP after the RD met); close it
```

13.3 Why each gap

- Read-to-write turnaround: the write command may be issued no earlier than RL + RU(tDQSCKmax/tCK) + BL/2 + 1 - WL clocks after the read command. With RL=3, WL=1, BL=4 and tDQSCKmax fitting in 2 clocks that is 3 + 2 + 2 + 1 - 1 = 7 clocks. The bubble exists because read data is still leaving the DRAM (tDQSCK is long and unpipelined - there is no DLL) and the DQ bus must be idle before the controller starts driving write data.
- Write-to-read: the next read command waits WL + 1 + BL/2 + RU(tWTR/tCK) after the write command (1 + 1 + 2 + RU(7.5 ns/tCK) here). tWTR lets the write data finish propagating into the array before sense amps are asked to read.
- `RD B1 C4` - same bank, same row as the write: a row hit even though a write happened in between. Open-row state is per bank, not per access type.
- `PRE B1` - legal once tRAS (from the ACT that opened R2) and tRTP (from RD B1 C4) are both met.

13.4 AXI view

AXI is oblivious to direction changes; its read and write channels are independent. The DRAM bus is not: every RD->WR or WR->RD transition on the shared DQ pins costs turnaround clocks, so a controller that groups reads with reads and writes with writes (batching) measurably outruns one that serves requests strictly in arrival order.

13.5 What to notice

- Turnarounds, not tRCD, dominate lightly-loaded mixed traffic.
- LPDDR2 has no ODT to switch and no tRTW parameter - the turnaround rules are formulas over RL, WL, BL and tDQSCK, all given in Chapter 6.
- tCCD still applies between the two column commands: even without the direction change, bursts on the shared bus must not collide.

Source: JESD209-2F sections 5.4, 5.5, 5.9.1 (derived walkthrough)

14 Data Path

14.1 Prefetch

- S4: 4n prefetch. One column access reads or writes 4 words per DQ inside the array; the interface serializes them as DDR data over 2 clock cycles (minimum burst BL4 = 2 clocks on the bus).
- S2: 2n prefetch. The minimum burst occupies 1 clock (BL4 = 1 clock on the bus), which is why tCCD = 1 on S2 and 2 on S4.

The prefetch is what lets a 533 MHz / 1066 MT/s interface coexist with a core that runs a fraction of that speed.

14.2 Burst ordering

MR1 programs BL (4/8/16), burst type and wrap control:

Mode	Behavior
Sequential + wrap	Addresses count up from the start column and wrap at the BL boundary
Interleaved + wrap	Two-word pairs swap order within the BL block (SDRAM only)
No-wrap	BL4 only: 4 words count straight up, may not cross page/sub-page boundaries

Wrap exists for cache-line fills: the critical word comes first and the rest of the line follows regardless of starting offset. BL16 with interleaved order is not a legal combination.

14.3 DQS: the bidirectional strobe

- One differential pair (DQS_t/DQS_c) per byte lane; x16 has 2 pairs, x32 has 4.
- Reads: the DRAM drives DQS edge-aligned with DQ, preceded by a tRPRE low preamble and followed by a tRPST postamble. The controller captures DQ on both DQS edges.
- Writes: the controller drives DQS center-aligned in the DQ eye, with a tWPRE preamble and tWPST postamble; the DRAM samples DQ on both DQS edges (tDS/tDH around each edge).
- There is no DLL. Read timing is bounded by tDQSCK (2.5-5.5 ns from the clock) and tDQSCK may span multiple clock periods - the controller must train its capture logic (e.g. with the MR32/MR40 DQ calibration patterns) rather than assume DQS is clock-aligned.

14.4 DM - write data mask

One DM pin per byte lane, sampled on both DQS edges with the write data: DM high masks that beat's byte from being written. DM is how a controller implements partial writes (AW/ W with WSTRB gaps) without read-modify-write. DM loading must match DQ/DQS loading. The DNV (data-not-valid) variant of the pin is an NVM feature; LPDDR2 SDRAM does not implement it and does not drive the pin.

14.5 No DBI, no parity, no ECC

LPDDR2 has none of the data-integrity helpers of later standards:

- No DBI (data bus inversion) on reads or writes.
- No parity on the CA bus and no CRC on the data bus.
- No on-die ECC.

Every bit on DQ is exactly what the controller put there or exactly what the array holds. If the system needs error detection or correction it must be done in the controller (ECC over the data it writes), in the interconnect, or not at all. This also means the DQ bus carries data only - never sideband metadata - so every pin cycle of the bus is payload.

14.6 Termination and signaling

The IO is HSUL_12: unterminated, 1.2 V VDDQ, VREFDQ = VDDQ/2 at the receiver. There is no ODT pin and no termination to switch - which is why bus turnaround rules are simple contention-avoidance formulas (Chapter 6) rather than ODT timing tables. Output drive strength is set in MR3 and maintained by ZQ calibration against an external 240 ohm resistor (S4/N only).

Source: JESD209-2F sections 2.12, 3 (functional description), 3.5 (Table 21), 5.4-5.7, 9

15 Core Timing

Core parameters govern row and column command spacing within and between banks. Values below are the spec's ns bins (fast / typical / slow), which apply across speed grades unless noted; each parameter also has a minimum in clocks (min tCK). RL/WL and tDQSCK are frequency-dependent.

15.1 Row-cycle parameters

Symbol	Definition	Applies between	Min (typ bin)
tRCD	RAS-to-CAS delay: row open to first column command	ACT -> RD/WR to same bank	15 / 18 / 24 ns, min 3 tCK
tRAS	Row active time: minimum time a row must stay open	ACT -> PRE to same bank	42 ns, min 3 tCK; max 70 us
tRPpb	Precharge time, single bank	PRE (one bank) -> ACT to that bank	15 / 18 / 24 ns, min 3 tCK
tRPab	Precharge time, all banks	PRE-all -> ACT to any bank	4-bank: = tRPpb; 8-bank: 18 / 21 / 27 ns, min 3 tCK
tRC	Row cycle: minimum ACT-to-ACT to the same bank	ACT -> ACT, same bank	tRAS + tRPpb (or tRAS + tRPab after PRE-all)
tRRD	ACT-to-ACT spacing, different banks	ACT -> ACT, different banks	10 ns, min 2 tCK
tFAW	Four-activate window: at most 4 ACTs in any rolling window	1st -> 5th ACT (8-bank only)	50 ns (60 ns slowest grade), min 8 tCK

tRPab exceeds tRPpb only on 8-bank devices - it keeps an all-bank precharge inside the current envelope of a 4-bank part. REFpb counts as an activation for tFAW.

15.2 Column-cycle parameters

Symbol	Definition	Applies between	Min (typ bin)
tRTP	Read to precharge: last-prefetch analog delay before PRE	RD -> PRE to same bank	7.5 ns, min 2 tCK
tWR	Write recovery: last write data to PRE	end of WR burst -> PRE to same bank	15 ns, min 3 tCK
nWR	tWR expressed in clocks for auto-precharge	(MR1 field)	RU(tWR/tCK), 3-8

For tRTP the spec's clock counting starts BL/2 - 2 clocks after the RD command on S4 (BL/2 - 1 on S2); the 7.5 ns analog floor then applies from the last internal prefetch. With AP=1 the device enforces these itself - the controller only has to have programmed nWR correctly.

15.3 Latency programming (MR2) and read access time

Speed grade	RL	WL	tCK (ns)
LPDDR2-1066	8	4	1.875
LPDDR2-933	7	4	2.15
LPDDR2-800	6	3	2.5
LPDDR2-667	5	2	3.0
LPDDR2-533	4	2	3.75
LPDDR2-466 and slower	3	1	>= 4.3

Symbol	Definition	Applies between	Value
tDQSCK	Strobe output access time from clock	RD -> DQS/DQ valid	2500-5500 ps (may span multiple clocks)
tDQSQ	DQS-to-DQ skew within a beat	DQS edge -> DQ edge	200 ps (1066) to 700 ps (200)
tQHS	Data hold skew factor	hold-time budgeting	230 ps (1066) to 1000 ps (200)

First read data arrives RL*tCK + tDQSCK + tDQSQ after the RD command's clock edge. There is no DLL, so tDQSCK is an analog delay the controller must absorb with a trained capture, not a fixed clock count.

15.4 Temperature de-rating

When MR4 indicates the extended-temperature de-rating code, add 1.875 ns to tRCD, tRC, tRAS, tRP (both variants) and tRRD, and use the de-rated tDQSCK (up to 5620/6000 ps). Clock frequency and setup/hold timings do not change.

Source: JESD209-2F sections 5.1, 5.4-5.5, 5.9, 12.4 (Table 103)

16 Turnaround and Burst Timing

Turnaround parameters govern the shared DQ bus when direction or burst changes. LPDDR2 has no ODT and no DLL, so these are contention and analog-settling rules, expressed as formulas over RL, WL, BL and tDQSCK rather than as a single named constant per transition.

16.1 Burst-to-burst spacing

Symbol	Definition	Applies between	Min
tCCD	Column-to-column command spacing	RD -> RD, or WR -> WR (any banks)	S4: 2 tCK; S2: 1 tCK

Interruption rules: on S4 a BL4 burst may never be interrupted; with BL8/BL16 a RD may interrupt a RD and a WR may interrupt a WR only on even clock cycles after the first command, tCCD met. On S2 bursts may be interrupted whenever tCCD is met. RD <-> WR interruption is not allowed in either class; use BST first if a burst must be cut short.

16.2 Write-to-read turnaround

Symbol	Definition	Applies between	Min
tWTR	Internal write-to-read delay	end of WR burst data -> RD	7.5 ns (1066-533) / 10 ns (466-200), min 2 tCK

The schedulable form, from the WR command to the RD command:

```
WR -> RD  >=  WL + 1 + BL/2 + RU(tWTR/tCK)   clocks
```

(BL is the effective burst length if the write was truncated by BST.)

16.3 Read-to-write turnaround

JESD209-2F gives the read-to-write rule as a formula without assigning it a symbol. This book calls it tRTW because the other specifications in this series name that parameter, and a common symbol set keeps the timing tables comparable across books.

Symbol	Definition	Applies between	Min
tRTW	Read-to-write turnaround (book symbol; the spec gives the rule unnamed, as a formula)	RD(any bank) -> WR(any bank)	RL + RU(tDQSCKmax/tCK) + BL/2 + 1 - WL clocks

The terms, in words: wait out the read latency and the (possibly multi-clock) strobe access time, let the burst finish and one bubble clock pass, then credit back the write latency because write data is timed from the WR command itself. With RL=3, WL=1, BL=4, RU(tDQSCKmax/tCK)=2: 7 clocks.

16.4 Burst terminate

BST truncates the most recent RD or WR. Effective burst length after BST = 2 x (clocks from RD/WR to BST), and that effective BL feeds every formula above. Issue window: up to BL/2 - 1 clocks after the RD/WR on S4; any cycle after the command on S2. The truncation lands one full RL (reads) or WL (writes) later.

16.5 MRR placement

The mode-register read is a BL4 read-like operation on the DQ bus, so it obeys the same spacing: no earlier than BL/2 after a RD, and no earlier than WL + 1 + BL/2 + RU(tWTR/tCK) after a WR. MRR to a following write: RL + RU(tDQSCKmax/tCK) + 4/2 + 1 - WL clocks.

16.6 Example gap sheet (RL=3, WL=1, BL=4, S4, 3 ns tCK)

Transition	Gap (clocks)
RD -> RD	2 (tCCD)
WR -> WR	2 (tCCD)
RD -> WR	7
WR -> RD	1 + 1 + 2 + 3 = 7 (RU(7.5/3) = 3)
RD -> PRE (same bank)	BL/2 - 2 = 0 clocks + tRTP 7.5 ns analog
WR -> PRE (same bank)	end of data + tWR 15 ns

Source: JESD209-2F sections 5.3-5.6, 5.9.1-5.9.2, 5.12, 12.4 (Table 103)

17 Refresh Timing

LPDDR2 gives the controller unusual freedom in WHEN refreshes happen: a count of refresh commands owed per rolling window, two granularities (all-bank and per-bank), and a temperature-scaled rate. This file is the complete picture.

17.1 The contract

Symbol	Definition	Value
tREFW	Rolling refresh window: R refreshes must land in every window	32 ms (Tcase <= 85 C); 8 ms (85-105 C)
R	Required REFab count per tREFW	2048 (64-128 Mb), 4096 (256 Mb-1 Gb), 8192 (2-8 Gb)
tREFI	Average all-bank refresh interval (= tREFW / R, reference only)	15.6 / 7.8 / 3.9 us by density
tREFIpb	Average per-bank refresh interval (reference only)	0.975 us (1 Gb S4), 0.4875 us (2-8 Gb)
tRFCab	All-bank refresh cycle time: device busy	90 ns (<=1 Gb), 130 ns (2-4 Gb), 210 ns (6-8 Gb)
tRFCpb	Per-bank refresh cycle time: target bank busy	60 ns (<=4 Gb), 90 ns (6-8 Gb)
tREFBW	Burst refresh window: at most 8 REFab in any rolling window of this length	4 x 8 x tRFCab (2.88 / 4.16 / 6.72 us)

One REFab may be replaced by a full round of eight REFpb (which is why tREFIpb = tREFI / 8). REFpb exists only on 8-bank devices (1 Gb S4 and up); S2 devices are REFab-only below 4 Gb.

17.2 Scheduling freedom, and its limits

- Regular pattern: one REFab every tREFI (or one REFpb every tREFIpb). Self-refresh may be entered at any time.
- Burst/pause pattern: up to 8 REFab per rolling tREFBW, then silence until the window count demands more. In the extreme, an entire window's R refreshes can be bursted, leaving up to tREFW - R x 4 x tRFCab refresh-free (about 30 ms for a 1 Gb S4).
- Any pattern is legal if EVERY rolling tREFW contains R refreshes and every rolling tREFBW contains at most 8 REFab. Transitions between patterns are where violations happen: entering self-refresh is only safe directly after a burst phase, because self-refresh behaves like the regular distributed pattern inside the window math.

17.3 Self-refresh interaction

Time in self-refresh earns credit: the required count in a window containing tSRF of self-refresh drops to R* = R - RU(tSRF / tREFI). After exit (tXSR = tRFCab + 10 ns), at least one refresh (1 REFab or a full round of 8 REFpb) must be issued before the next self-refresh entry, because an internal refresh may have been cut short at exit.

17.4 REFpb mechanics

- The target bank is chosen by the device's round-robin counter, synchronized to bank 0 at reset, at every self-refresh exit, and at every REFab. The controller must shadow this counter.
- During tRFCpb only the target bank is busy; all other banks may be activated, read or written. REFpb also counts as a bank activation for tFAW.
- Separations: tRFCpb after a REFpb before ACT-to-same-bank or the next REF; tRFCab after REFab before anything; only tRRD before an ACT to a different bank than the REFpb target; tRP after a PRE before refreshing that bank.

17.5 PASR - partial array self-refresh

In self-refresh, masked regions are simply not refreshed (and lose their data), cutting standby current:

- S2: MR16 picks full / 1/2 / 1/4 / 1/8 array, anchored at bank 0.
- S4: MR16 is a per-bank mask; MR17 (1 Gb and up) adds a per-segment mask over 8 row-space segments. Bank and segment masks combine: a location is refreshed only if its bank and segment are both unmasked.

17.6 Temperature-compensated self refresh (TCSR)

The device reads its own temperature and posts a recommended refresh rate in MR4 (4x / 2x / 1x / 0.25x tREFI, with a de-rate flag at the high end). The controller polls MR4 - the internal sensor updates at most every tTSI = 32 ms, and the read interval must satisfy TempGradient x (ReadInterval + tTSI + SysRespDelay) <= 2 C - and scales its refresh rate (and, above 85 C, de-rates core AC timings by 1.875 ns). Below 85 C the 0.25x option roughly quarters refresh power.

Source: JESD209-2F sections 5.10-5.12.1, 12.3 (Tables 101-102), 12.4 (Table 103)

18 Package and Electrical Notes

Light notes on the physical side; the spec's package section is primarily ballout drawings, which are not reproduced here.

18.1 Supplies

Rail	Typ	Feeds
VDD1	1.8 V	Core (low-current domain)
VDD2	1.2 V (S4B; 1.35 V S4A; absent on S2A)	Core (main current)
VDDCA	1.2 V	CA/CKE/CS_n/CK input receivers
VDDQ	1.2 V	DQ/DQS/DM IO buffers
VREFCA / VREFDQ	rail / 2	Input reference for CA and DQ receivers
ZQ	-	External 240 ohm +/- 1% calibration reference

Ramp-order rules (Chapter 2) keep VDD1 within 200 mV above VDD2, and both within 200 mV above VDDCA/VDDQ, at all times. VDDQ may be switched off in power-down and self-refresh (VREFDQ must follow it), which is a real system-level power lever.

18.2 Interface: HSUL_12

LPDDR2's IO is high-speed unterminated logic at 1.2 V: point-to-point or short multi-drop, no termination resistors, no ODT. Drive strength is programmed in MR3 (34-120 ohm typical options, default 40 ohm) and held over voltage/temperature by ZQ calibration (tZQINIT at boot, tZQCL/tZQCS maintenance commands thereafter). Clock and strobes are differential (CK_t/CK_c, DQS_t/DQS_c); everything else is single-ended.

18.3 Packages

The spec defines BGA ballouts for both PoP (package-on-package, stacked under the SoC) and discrete/MCP use, including:

- 168-ball 12x12 mm PoP, 2-channel 2x32 (two independent x32 channels plus shared power in one package)
- 134-ball and 162/180-ball x16/x32 discrete/MCP options
- 79-ball x16 small-pitch discrete
- MCP ballouts pairing LPDDR2 with SDR NAND or e-MMC on one package

Multi-channel packages give each channel its own CA bus, CK, CKE and CS_n - they are independent memories sharing a package and nothing else.

18.4 Power-state current hierarchy (why the states exist)

Roughly, from most to least expensive: active operation > active power-down > idle power-down > self-refresh (with PASR reducing it further) > deep power-down (contents lost) > powered off. The CKE table in Chapter 3 is the map between them.

Source: JESD209-2F sections 2 (package ballouts), 6-7 (Tables 69-73), 9, 11

19 Test Features

LPDDR2 is a pre-boundary-scan mobile part: there is no JTAG boundary scan, no OCD calibration engine like DDR2's, and no post-package repair in the standard. What it does have:

19.1 DQ calibration patterns (MR32 / MR40)

Two read-only mode registers return fixed, known data patterns on the DQ bus. The controller issues an MRR to MR32 or MR40 and uses the known pattern to train its read capture - essential because LPDDR2 has no DLL and tDQSCK is an analog window of 2500-5500 ps rather than a fixed clock count. Unlike a normal MRR (valid data on the first beat only), the DQ calibration MRRs define content across the burst so timing can be tuned on repeated edges.

19.2 ZQ calibration (MR10)

Output-driver self-test and calibration against the external 240 ohm ZQ resistor: initialization calibration (0xFF) at boot, long (0xAB) and short (0x56) maintenance calibrations, and ZQ reset (0xC3) for default calibration. MR0's RZQI field reports the ZQ pin connection self-test result (floating/shorted/normal) after the initialization calibration. S2 devices ignore all ZQ commands.

19.3 Temperature sensor (MR4)

Covered in Chapters 2 and 6. From a test perspective it is also the only on-die “measurement” the host can read: a 3-bit refresh-rate recommendation plus the TUF change flag, updating no faster than tTSI = 32 ms.

19.4 Vendor test mode (MR9)

A write-only register reserved for vendor-specific test modes. The spec defines no behavior; using it outside a vendor datasheet is undefined.

19.5 Identity registers

MRR of MR0 (device info, DAI status), MR5 (manufacturer ID), MR6/MR7 (revision IDs) and MR8 (type, density, IO width) is the standard bring-up sanity check: if the values decode sensibly, the CA interface, the MRR path and the read capture are all working.

Source: JESD209-2F sections 3.5 (MR9, MR10, MR32/MR40), 5.12.2, 5.13.4